

Review of postcontrast MRI studies on diffusion of human lumbar discs.

Diffusion is the only source of nutrition to the intervertebral discs, and alteration of diffusion is considered to be the final common pathway for disc degeneration. Yet diffusion remains poorly understood due to the paucity of reliable methods to study diffusion noninvasively in humans in vivo. In recent years, postcontrast MRI has emerged as a powerful and reliable tool for analyzing diffusion in lumbar discs. Since it is noninvasive and safe, it can be used to document the process of diffusion temporally over a period of 24 hours. Well-designed studies have shown that diffusion is a very slow process, and that the endplate is the main structure that controls the process of diffusion. Contrast MRI studies have also made it possible to identify endplate breaks in vivo. In the future this technique may be applied to study the influence of smoking, mechanical loading of the discs, abnormal posture, and atherosclerosis of the lumbar arteries on diffusion. These conditions have all been implicated in disc degeneration through a final common pathway of altered diffusion and decreased nutrition. This review article focuses on the current knowledge, methodology, various factors that influence the diffusion properties of the discs, and future applications of this promising technique. (c) 2007 Wiley-Liss, Inc.